At temperature $T = T_2$; the rate constant $k = k_2$

$$\ln k_2 = \ln A - \left(\frac{E_a}{RT_2}\right)$$

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$$\ln k_2 - \ln k_1 = -\left(\frac{E_a}{RT_2}\right) + \left(\frac{E_a}{RT_1}\right)$$

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$2.303 \log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\ln k_2 - \ln k_1 = -\left(\frac{E_a}{RT_2}\right) + \left(\frac{E_a}{RT_1}\right)$$

This equation can be used to calculate E_a from rate constants k_1 and k_2 at temperatures T_1 and T_2.

Example 7

The rate constant of a reaction at 400 and 200K are 0.04 and 0.02 s^{-1} respectively. Calculate the value of activation energy.

Solution

According to Arrhenius equation

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$T_2 = 400K$; $k_2 = 0.04\ s^{-1}$

$T_1 = 200K$; $k_1 = 0.02\ s^{-1}$

$$\log\left(\frac{0.04\ s^{-1}}{0.02\ s^{-1}}\right) = \frac{E_a}{2.303 \times 8.314\ JK^{-1}mol^{-1}}\left(\frac{400K - 200K}{200K \times 400K}\right)$$

$$\log(2) = \frac{E_a}{2.303 \times 8.314\ JK^{-1}mol^{-1}}\left(\frac{1}{400K}\right)$$

$E_a = \log(2) \times 2.303 \times 8.314\ JK^{-1}mol^{-1} \times 400K$

$E_a = 2305\ J\ mol^{-1}$

Example 8

Rate constant k of a reaction varies with temperature T according to the following Arrhenius equation

$$\log k = \log A \frac{E_a}{2.303R}\left(\frac{1}{T}\right)$$

Where E_a is the activation energy. When a graph is plotted for $\log k$ Vs $\frac{1}{T}$ a straight line with a slope of -4000K is obtained. Calculate the activation energy

Solution

$$\log k = \log A - \frac{E_a}{2.303R}\left(\frac{1}{T}\right)$$

$y = c + mx$

$$m = -\frac{E_a}{2.303R}$$

$E_a = -2.303\ R\ m$

$E_a = -2.303 \times 8.314\ J\ K^{-1}\ mol^{-1} \times (-4000K)$

$E_a = 76,589 J\ mol^{-1}$

$E_a = 76.589\ kJ\ mol^{-1}$

Evaluate yourself

For a first order reaction the rate constant at 500K is $8 \times 10^{-4}\ s^{-1}$. Calculate the frequency factor, if the energy of activation for the reaction is 190 kJ mol^{-1}.